

Agustín Avelino Pineda

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PROFESSIONAL PROFILE

Mechatronics Engineering student with competencies in embedded systems, automatic control, power electronics, analog instrumentation and PCB design, supported by C++ and Python programming and CAD modeling. I bring technical leadership, collaborative teamwork and problem solving under demanding deadlines. My professional interest is focused on automation, automotive systems and mobile robotics.

EDUCATION

Tec de Monterrey Campus Santa Fe

B.S. in Mechatronics Engineering

Santa Fe, Mexico City

Expected graduation: August 2028

CECyT 9 “Juan de Dios Bátiz” — Instituto Politécnico Nacional

Technical Degree in Computer Programming

Mexico City

March 2025

PROFESSIONAL EXPERIENCE

Tec de Monterrey Campus Santa Fe

Line-Following AGV with Load/Unload System — Engineering Project

Santa Fe, Mexico City

February – June 2026

- Designed and implemented a PD controller with dead zone and exponential smoothing filter on the PWM signal, achieving stable autonomous navigation on a differential-drive robot with BTS7960 drivers.
- Developed line tracking, path-loss recovery and station detection in C++, through more than 20 test iterations.
- Integrated an HX711 load cell with a Kalman filter, encoders (494 PPR), a TMC2209 stepper motor and a 400 kHz I2C bus with PCF8574 and OLED, achieving stable weighing while in motion.

Hybrid Solar/Electric Food Dehydrator — Industry Partner Project

August – September 2026

- Designed the analog instrumentation and control system of a food dehydrator with two energy sources (solar collector and 127 VAC heating element), without microcontrollers or programmable devices.
- Conditioned two process variables: temperature with a PT100 in a Wheatstone bridge and an AD620 instrumentation amplifier (20–100 °C, ratiometric), and relative humidity with a capacitive sensor, characterizing its systematic temperature error.
- Developed the schematic and two-layer PCB layout in KiCad (74 components): net classes, ground plane, separation of switching and measurement zones, test points and calibration silkscreen.
- Validated both measurement chains on a breadboard and diagnosed three faults against datasheet specifications.

WORK EXPERIENCE

Omega Lightning — Student Racing Team, Tecnológico de Monterrey

Lead, Electronics Division

Santa Fe, Mexico City

February – August 2026

- Led the development of the electrical system of a 51.2 V / 300 A peak racing kart, specifying components and coordinating a team of 4 under competition deadlines.
- Resized the power wiring from 3/0 to 6 AWG using silicone insulation with higher thermal tolerance, reducing the weight of the electrical system.
- Designed (SolidWorks) and manufactured a sealed enclosure for the components and implemented the Safe-to-Touch system for chassis electrical isolation.
- Integrated solar panels for autonomous charging of the secondary electrical system, plus a speed-adaptive audio system.

Co-Lead, Electronics Division

August 2025 – December 2025

- Supported the initial integration of motor, batteries and lighting, laying the groundwork for the system I later led.

SKILLS

Programming: C++, Python, Git/GitHub, Linux, Arduino

Languages: Spanish (Native), English (Advanced)

Office: Excel (Intermediate)

CAD: SolidWorks, Siemens NX (Associate Certification)

Lab: Soldering, integration and troubleshooting

Electronics: PCB design in KiCad, signal conditioning